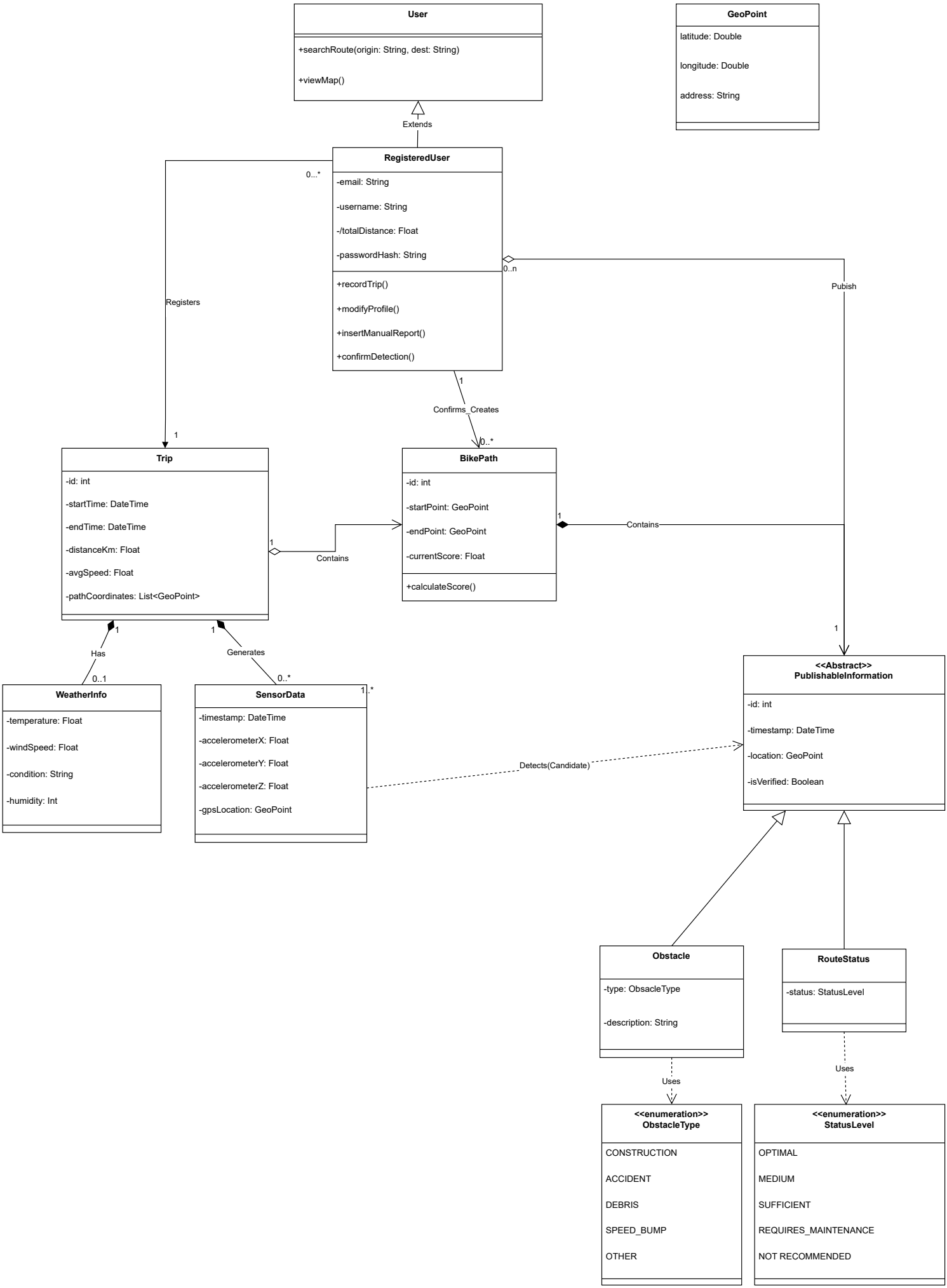
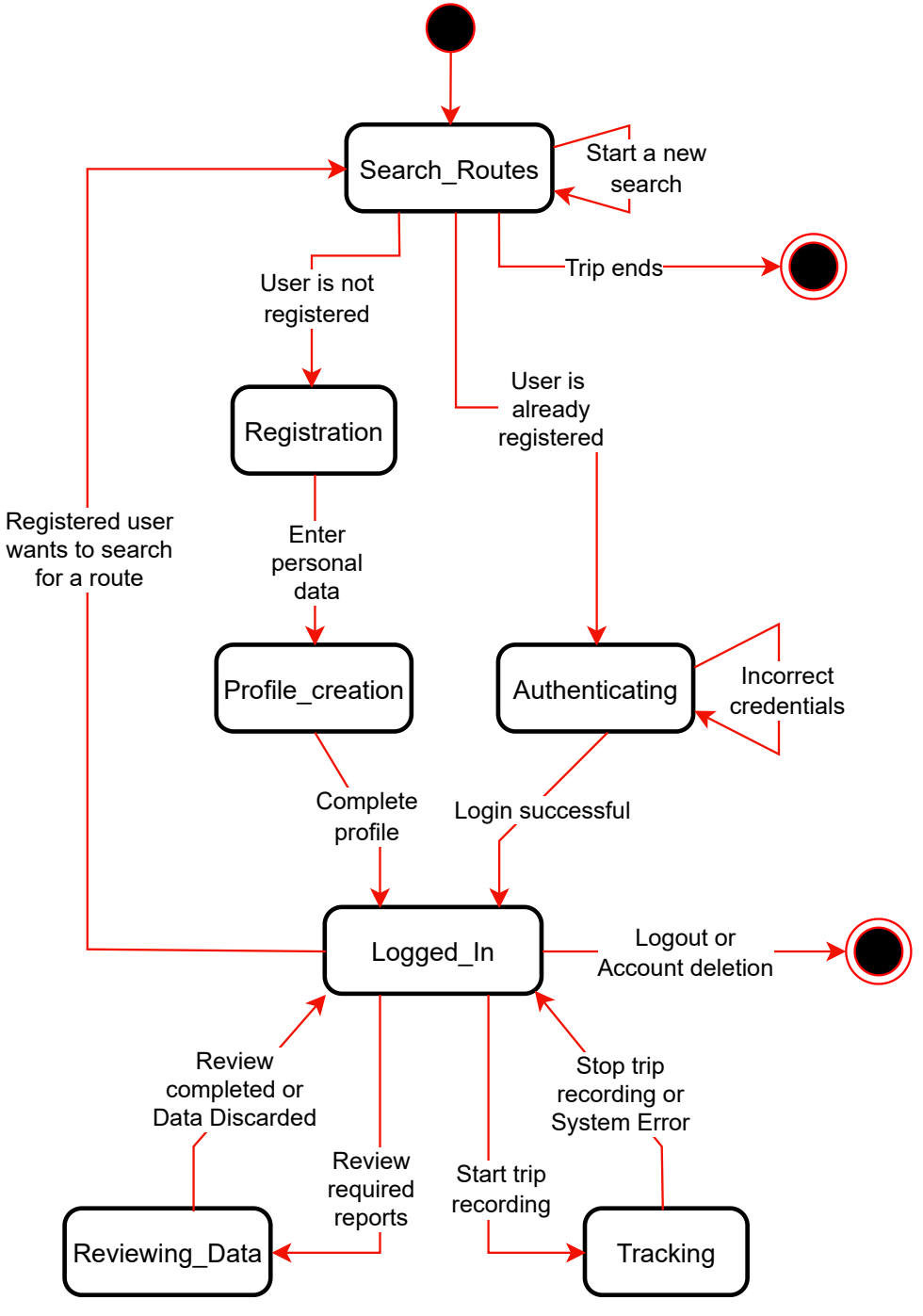
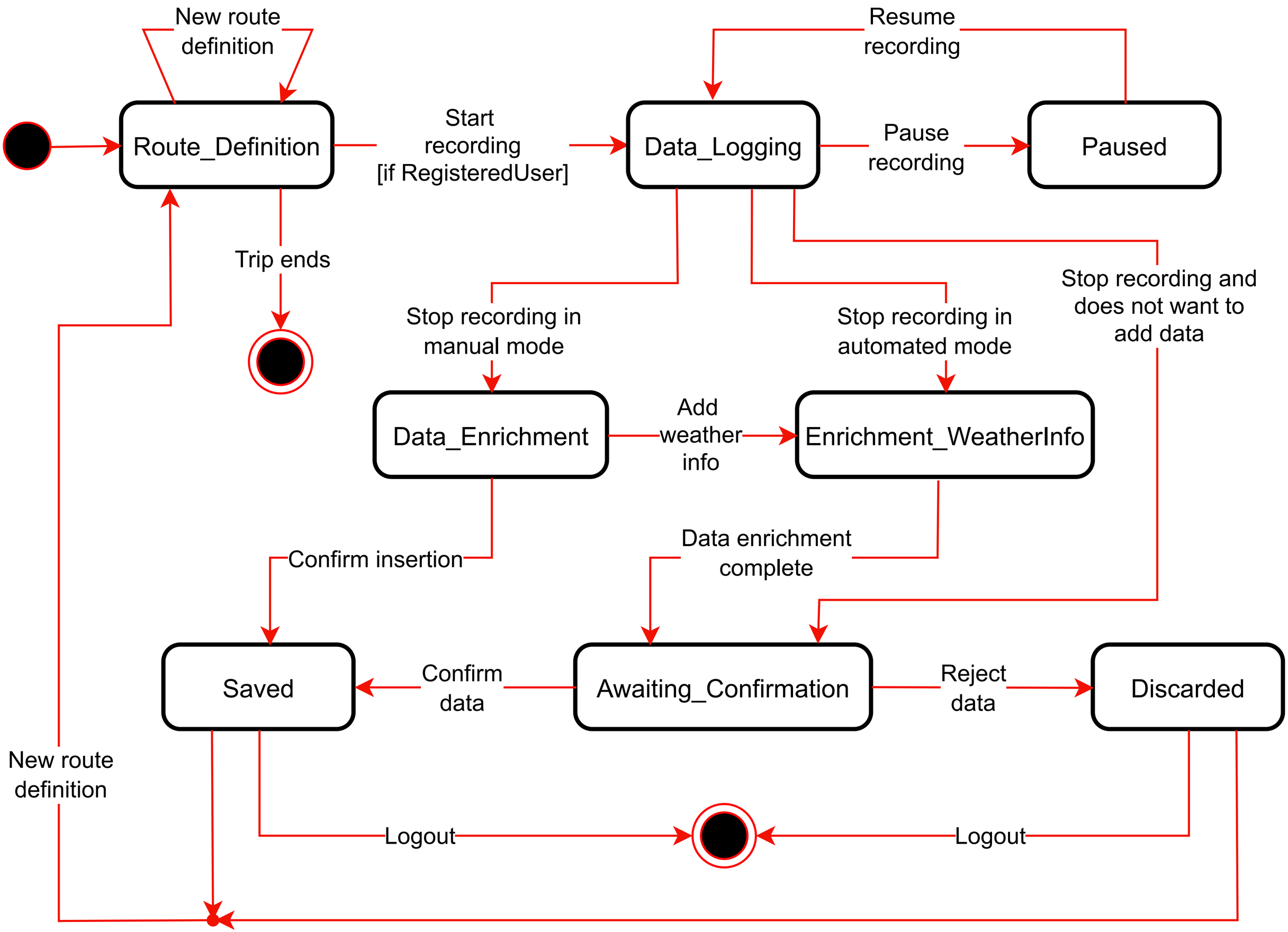
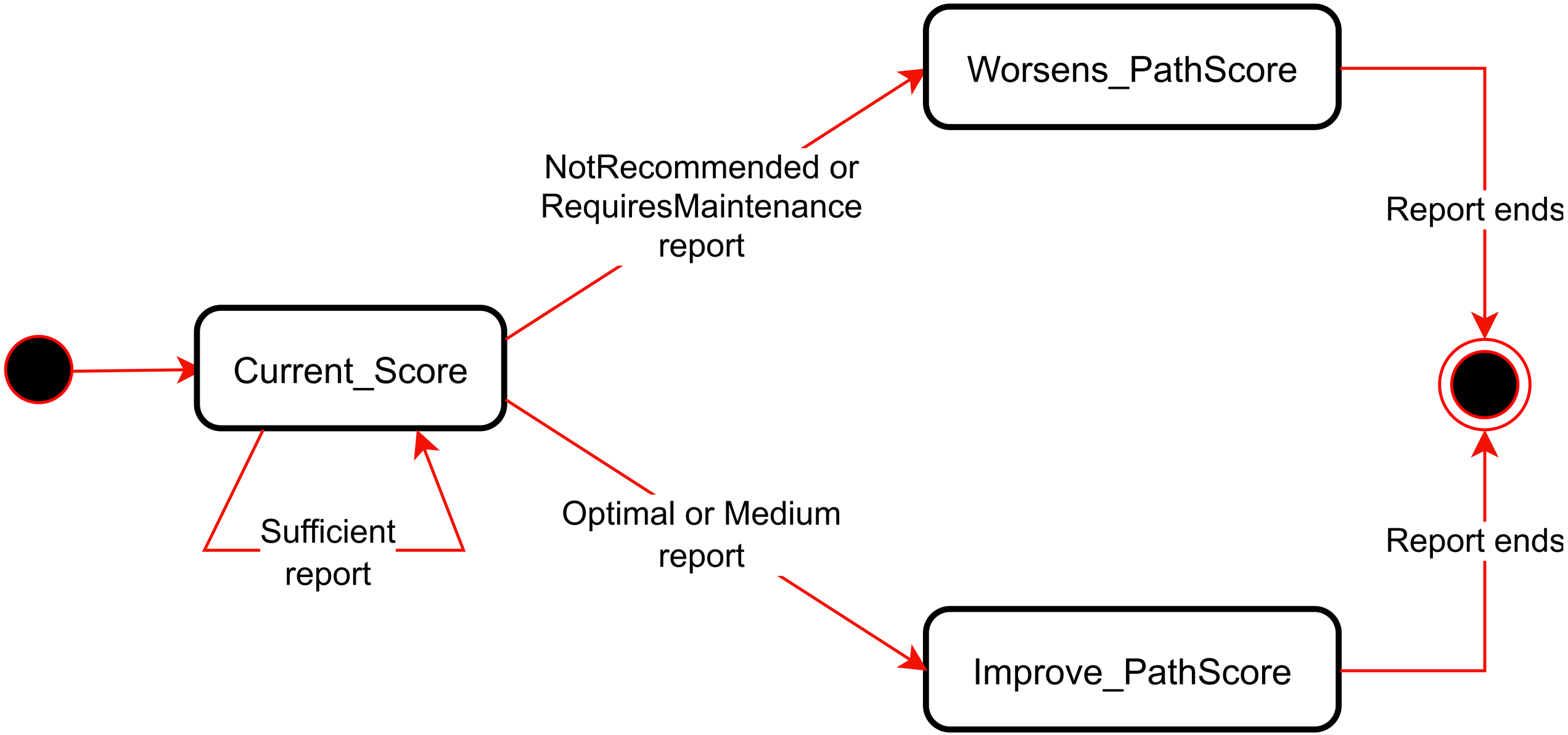
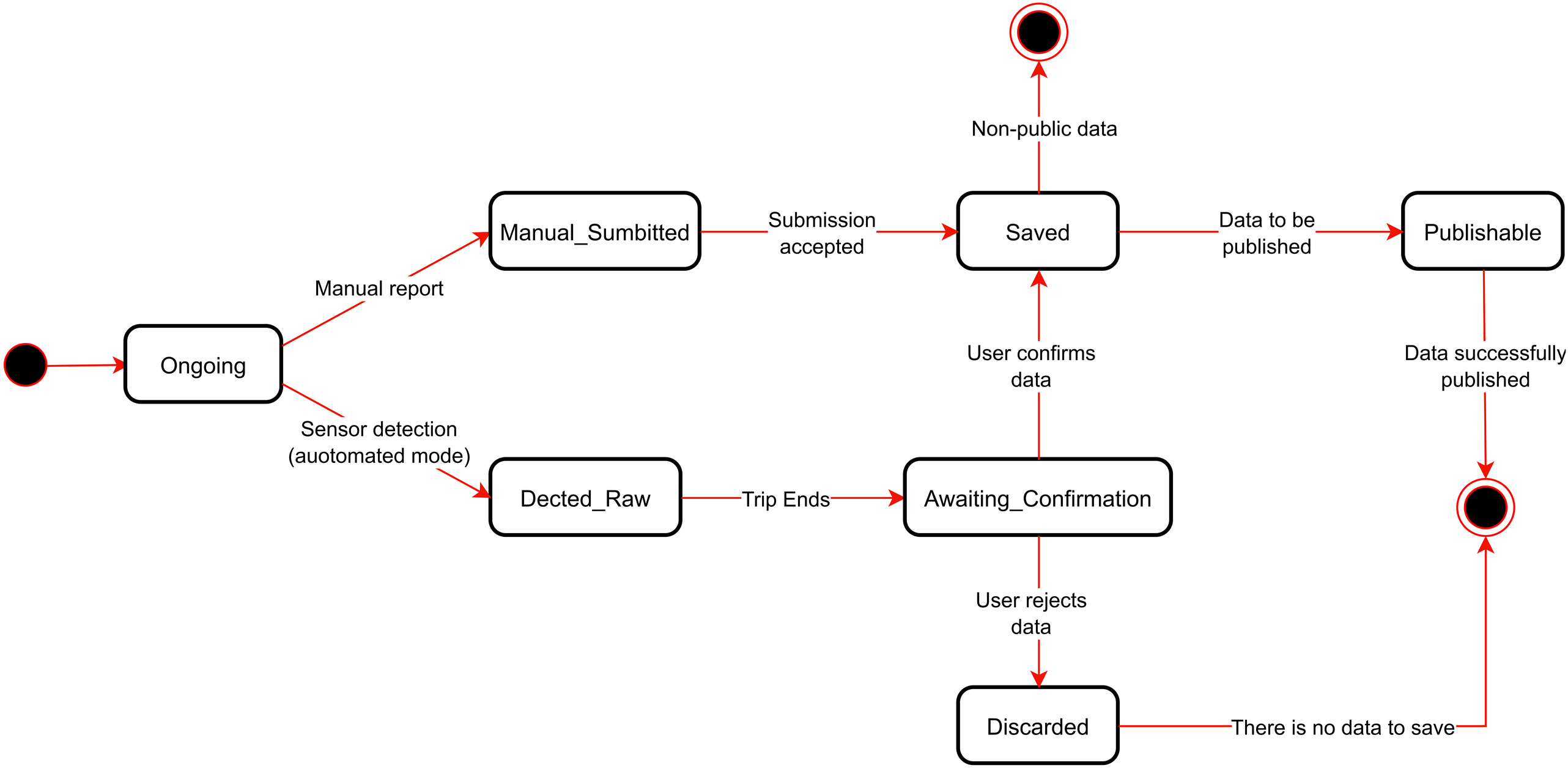

User
+searchRoute(origin: String, dest: String)
+viewMap()
GeoPoint
latitude: Double
longitude: Double
address: String
Extends
RegisteredUser
-email: String
-username: String
-/totalDistance: Float
-passwordHash: String
+recordTrip()
+modifyProfile()
+insertManualReport()
+confirmDetection()
0...*
Registers
0..n
Pubish
1
Confirms_Creates
0..*
Trip
-id: int
-startTime: DateTime
-endTime: DateTime
-distanceKm: Float
-avgSpeed: Float
-pathCoordinates: List<GeoPoint>
BikePath
-id: int
-startPoint: GeoPoint
-endPoint: GeoPoint
-currentScore: Float
+calculateScore()
Contains
Contains
Has
0..1
Generates
0..*
1..*
WeatherInfo
-temperature: Float
-windSpeed: Float
-condition: String
-humidity: Int
SensorData
-timestamp: DateTime
-accelerometerX: Float
-accelerometerY: Float
-accelerometerZ: Float
-gpsLocation: GeoPoint
Detects(Candidate)
<<Abstract>>
PublishableInformation
-id: int
-timestamp: DateTime
-location: GeoPoint
-isVerified: Boolean
Obstacle
-type: ObsacleType
-description: String
RouteStatus
-status: StatusLevel
Uses
Uses
<<enumeration>>
ObstacleType
CONSTRUCTION
ACCIDENT
DEBRIS
SPEED_BUMP
OTHER
<<enumeration>>
StatusLevel
OPTIMAL
MEDIUM
SUFFICIENT
REQUIRES_MAINTENANCE
NOT RECOMMENDED


Search_Routes
Start a new search
Trip ends
User is not registered
Registration
User is already registered
Registered user wants to search for a route
Enter personal data
Profile_creation
Authenticating
Incorrect credentials
Complete profile
Login successful
Logged_In
Logout or Account deletion
Review completed or Data Discarded
Review required reports
Start trip recording
Stop trip recording or System Error
Reviewing_Data
Tracking


New route definition
Route_Definition
Start recording [if RegisteredUser]
Data_Logging
Resume recording
Pause recording
Paused
Trip ends
Stop recording in manual mode
Stop recording in automated mode
Stop recording and does not want to add data
Data_Enrichment
Add weather info
Enrichment_WeatherInfo
Confirm insertion
Data enrichment complete
Saved
Confirm data
Awaiting_Confirmation
Reject data
Discarded
New route definition
Logout
Logout


Worsens_PathScore
NotRecommended or
RequiresMaintenance
report
Report ends
Current_Score
Sufficient
report
Optimal or Medium
report
Report ends
Improve_PathScore


Non-public data
Manual_Sumbitted
Submission accepted
Saved
Data to be published
Publishable
Manual report
Ongoing
User confirms data
Data successfully published
Sensor detection (auotomated mode)
Dected_Raw
Trip Ends
Awaiting_Confirmation
User rejects data
Discarded
There is no data to save